

BETTY WAIYEGO

Nairobi, Kenya | btyntambura20@gmail.com | +254727692909 | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Hands-on **AI Solutions Engineer** with proven experience rapidly prototyping, iterating, and deploying AI agents and automation tools. Skilled in **LLMs, LangChain, LangGraph, embeddings, and Python** for building scalable end-to-end solutions. Strong background integrating AI with business systems, vector databases, and serverless/cloud platforms (AWS, GCP). Adept at bridging business needs with technical solutions, delivering production-ready workflows that drive efficiency and impact.

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML, CSS

AI/ML & Automation: LLMs, VLMs, LangChain, LangGraph, AutoGen, Embeddings, Vector Search, Knowledge Graphs, Prompt Engineering

Cloud Platforms: AWS (Lambda, API Gateway, S3, Textract), GCP (Cloud Run, Compute Engine), Azure

Databases: Pinecone, MongoDB, SQL, NoSQL, Arango

Frameworks & Tools: FastAPI, Flask, Docker, Git, RESTful APIs

Practices: Agile/Scrum, CI/CD, Microservices, Rapid Prototyping & Iteration

PROFESSIONAL EXPERIENCE

AI Solutions Engineer

Prescott Data | Nairobi, Kenya | Mar 2025 – Present

- Partnered with leadership to identify high-impact use cases and rapidly deploy AI solutions into production.
- Built and deployed a **VLM-based OCR microservice** using FastAPI + Google Cloud Run with autoscaling.
- Designed and implemented **ReAct++ prompt engineering architecture** (instruction, action, reflection, evaluation layers).
- Prototyped and tested **Qwen2.5 7B VL model** for complex document and image extraction using ground-truth data.
- Developed a **production-ready chatbot** integrating Pinecone for knowledge retrieval, improving access to company policies by 90%.
- Implemented **embedding models** (AWS, DeepSeek) to boost chatbot accuracy and contextual relevance.
- Automated cost-saving by spinning up **GPU-enabled GCP VMs** with 30-min auto-shutdown logic.
- Built **serverless architectures** using AWS Lambda + API Gateway to ensure scalability and efficiency.
- Generated **Knowledge Graphs** for LLMs, enhancing contextual understanding.
- Deployed AI models inside AI Foundry and integrated into a VLM system.
- Used an OAuth broker deployed in Azure Container Apps to add provider apps for authenticating third-party users who need to access the apps.

- Configured CI/CD pipelines for deploying code to AKS and ACR.

AI Developer | TMA (Contract)

Jan 2023 – Feb 2025

Remote

- Built and deployed a **Vision-Language Model (VLM)** using **Mistral** models within **Azure AI Foundry**, enabling multimodal reasoning and contextual understanding from text and image inputs.
- Configured and optimized **Azure AI endpoints** for scalable model inference and evaluation pipelines.
- Took part in the development of a chatbot via Azure Bot Services and integrated it to MS365.
- Integrated the **VLM into internal AI workflows** to support use cases such as intelligent document parsing, data enrichment, and contextual retrieval.
- Designed and tested **prompt orchestration and model evaluation pipelines** using Azure AI Foundry's workflow tools.
- Collaborated on authentication and integration with **Azure AD** for secure API access and model endpoint protection.
- Developed **API reference documentation** for internal AI services, demonstrating usage of Foundry-deployed models within larger AI systems.

Tech Stack: Azure AI Foundry, Mistral, Azure AD, Python, REST APIs, React, JSON, Docker

EDUCATION

Bachelor of Science in Computer Science and Technology

Maseno University | Maseno, Kenya | 2013 – 2017

Software Engineering Program

ALX Africa | Mar 2024 – Jul 2025

PROJECTS

AI-Powered Chatbot Application

- Designed chatbot with LLMs + Python, integrated Pinecone for retrieval.
- Deployed via AWS Lambda + API Gateway for seamless scalability.
- Enhanced with embeddings for improved context.

Movie Recommendation System

- Built backend recommendation system in Python with real-time API integrations.
 - Deployed RESTful API endpoints with caching for optimized performance.
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CERTIFICATIONS

- AWS Cloud Practitioner (In Progress)
- ALX Software Engineering

PERSONAL QUALITIES

Rapid prototyper | Problem-solver | Clear communicator | Collaborative team player | Resilient and adaptable